An Automated Platform for GNSS Processing and Time-Series Analysis in a Complex Geodynamic Region

Dimitrios Anastasiou, Georgios Mytilinaios, Alexandros Mylonas, Georgios Serelis, Xanthos Papanikolaou, Vangelis Zacharis, Maria Tsakiri

Dionysos Satellite Observatory
School of Rural Surveying and Geoinformatics Engineering
National Technical University of Athens

http://dionysos.survey.ntua.gr/ danastasiou@mail.ntua.gr

Outline

- GNSS Networks in Greece
- Processing and Analysis Platform Upgrade
- Results & Outputs
- Time series analysis results
- Discussion / Conclusions

Motivation

Geodynamic context

Greece is one of the most active geodynamic regions in Europe, affected by tectonic deformation, strong earthquakes, and volcanic activity, such as in Santorini.

Routine GNSS processing and site/network monitoring allow us to:

- produce reliable and repeatable geodetic products for geodesy, seismology, volcanology, and broader Geoscience applications;
- estimate long-term station velocities and detect transient deformation, offsets, and post-seismic or volcanic signals;
- maintain processing strategies consistent with modern GNSS standards, reference frames, and EUREF/IGS practices;
- develop an automated platform for rapid, final, and high-rate GNSS solutions, supporting both research and academic services.

Key idea

Routine processing transforms heterogeneous GNSS observations into consistent time series and deformation products for a complex geodynamic region.

Data set and metadata challenges

Routine processing for precise positioning requires a well-established and reliable data set with complete metadata.

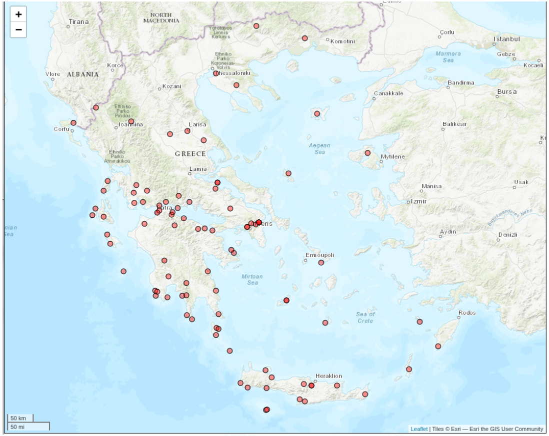
Currently we process whatever we can get our hands on ...
Problems:

- Inhomogeneous dataset (`RINEX` of various versions, raw files, etc).
- Various maintainers, different mentalities.
- Different acquisition methods/rates.
- No log files for maintainers with no geodetic interest (e.g. surveying companies).
- Wide variety of equipment (not always included in `ATX` files).

Network GREECE

Network GREECE includes the majority of the available sites (≈100) but not all of them are (always/currently) active. Various providers but all with geodetic interest & equipment.

- covers all of Greece
- different (geodetic type) equipment
- credible time-span (early 2004 - now)
- all freely available GNSS data
- large data gaps & inactive stations

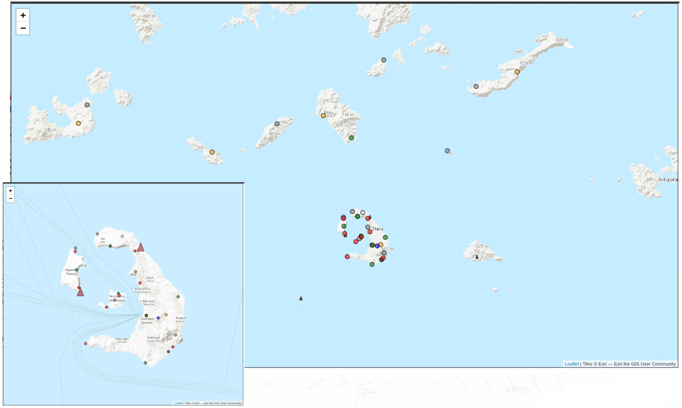


-» Network GREECE

Network SANTORINI

The **Santorini Volcano** ...

- credible time-span
- only covers the wider area of Santorini Volcano
- different providers (including surveying & cadastral services)
- no log files & equipment changes



-» Network Santorini

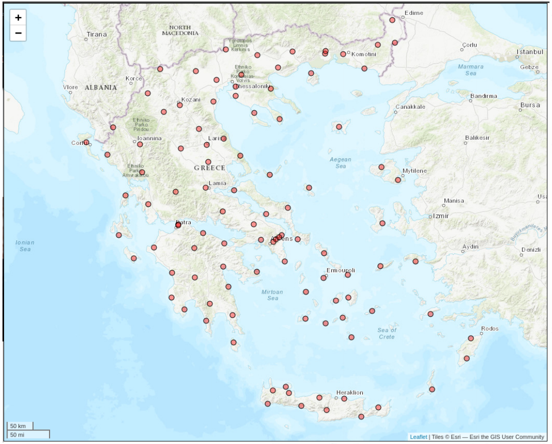
Network HEPOS

Network HEPOS

- covers all of Greece
- homogeneous equipment
- credible time-span (mid 2007 - now)
- only four years available for processing up to now

Data availability and funding from Hellenic Cadastre

HELLENIC CADASTRE



-» Network HEPOS

GNSS data processing and analysis at DSO

GNSS data analysis

- Bernese v5.4 (Dach, 2015)
- PRIDE-PPPAR (Geng et al., 2019)
- GipsyX (Bertiger et al., 2020)

Time series analysis

- Hector v2.1 (Bos et al., 2012)
- Python/C++ routines

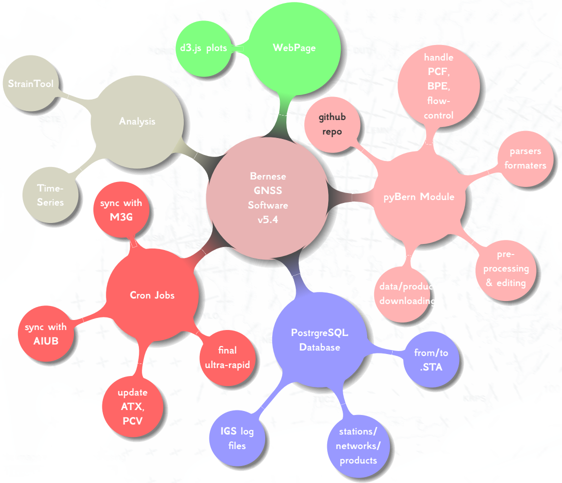
Solutions

- Daily solution
  - Rapid solution - 1 day after
  - Final solution - 12 days after
- High-rate analysis (>1 Hz)
  - data from network HEPOS - 1 Hz

Automation workflow

The core tool/software is `Bernese GNSS Software v5.4`.
Integration with

- **PostgreSQL** database,
- **Python** module (product/data downloading, pre-processing, driving cron jobs, etc) https://github.com/DSOlab/autobern
- **Time-series** analysis (integrated in routine processing on regular intervals)
- **Strain Rates** via StrainTool (on user demand)

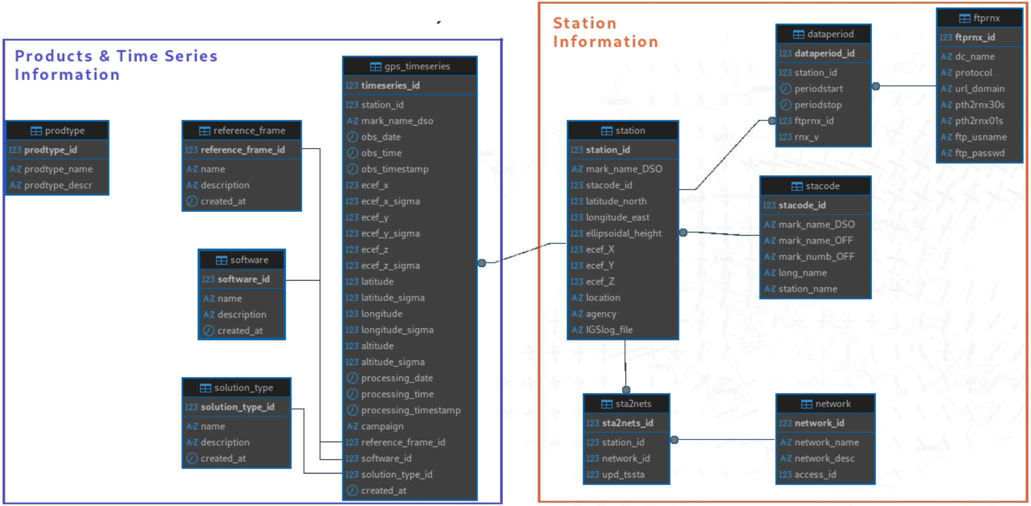


IGS20 transition for REPRO2026

- Reference frame `IGS20 > IGb20 > IGc20`
- Oceans loading corrections (`NTUA_FES2014b`)
- `VMF3` for tropospheric modeling
- `ATX` file: `igs20.atx`
- Long filename of products according to new IGS convention

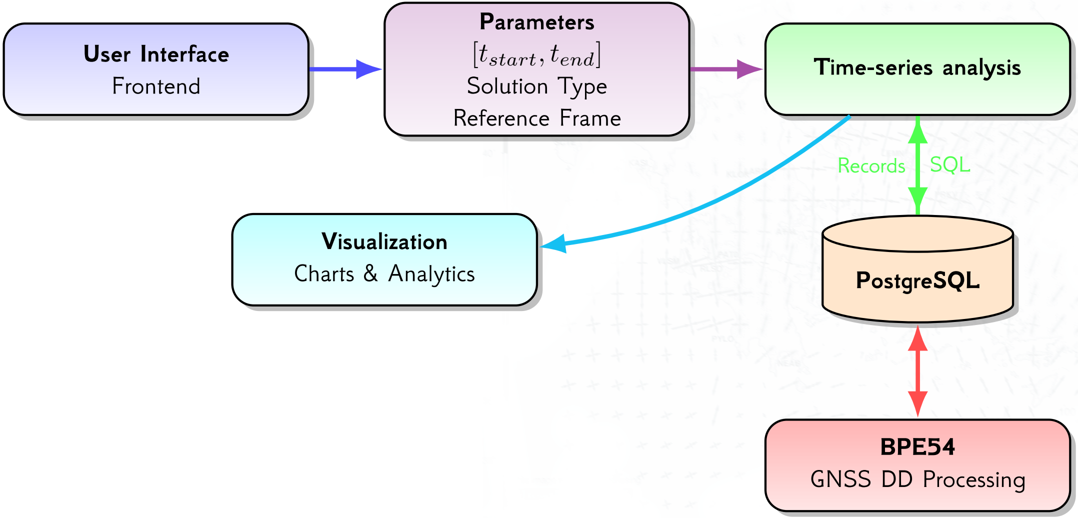
	Server	Start	Finish	REVIEW	UPLOAD EPN	start day	proc days	end day	Last script run	COMMENTS
1995										
1996										
...	...	...	...	...	...	...	...	...	...	...
2017	**bernese1**	**20.06.2026**		**ongoing**		**001**	**003**	**365**	**Sat Jun 20 10:45:47 EEST 2026**	
2018	**paraskevas**	**20.06.2026**		**ongoing**		**001**	**009**	**365**	**Sat Jun 20 09:50:14 EEST 2026**	**week=2008 start BIA, jul 1st/182**
2019	**dso01**	**20.06.2026**		**ongoing**		**001**	**025**	**365**	**Sat Jun 20 00:55:38 EEST 2026**	
2020	bernese1	13.06.2026		4REVIEW		001		366	FINISHED	
2021	paraskevas	12.06.2026		4REVIEW		001		365	FINISHED	
2022	dso01	12.06.2026	19.06.2026	4REVIEW		001		365	FINISHED	2022-331 -> change to new long format of products
2023	bernese1	05.06.2026	12.06.2026	FIN	OK ALL	001		365	FINISHED	
2024	paraskevas	02.06.2026	11.06.2026	FIN	OK ALL	001		366	FINISHED	
2025A	dso01	02.06.2026	04.06.2026	FIN	OK ALL	001		050	FINISHED	
2025B	dso01	06.06.2026	11.06.2026	FIN	OK ALL	051		365	FINISHED	switch to IGB20 02.02.2025
2026B	vincenty	16.06.2026	16.06.2026	FIN		001		010	FINISHED	
2026C	vincenty	16.06.2026	19.06.2026	4REVIEW		011		147	Tue Jun 16 11:54:16 EEST 2026	switch to IGC20 11.01.2026
2027										

Transition to the new PostgreSQL Database Schema



Time-series analysis ⇐ ⇒BPE54

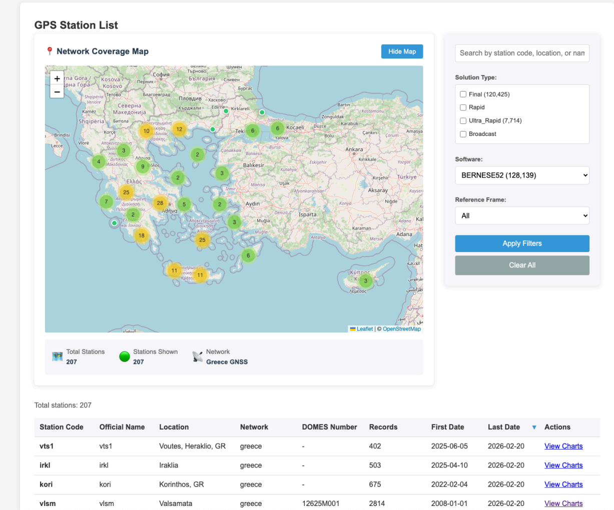
Platform workflow



Web Interface: station selection and filtering

- User entry point to the application.
- Selection of basic parameters:
  - ◇ GNSS station
  - ◇ Solution type
  - ◇ Software
  - ◇ Reference frame
- Filtering and reordering capability

Live app (beta version):



http://83.212.80.177:8585/

Processing outputs and products

4. Solution Indetifiers

Array of Objects

expand

5. PCF Variables

Array of Objects

expand

6. Saved products

Array of Objects

expand

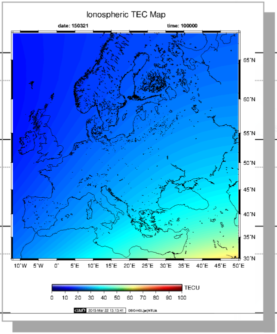
7. Warnings

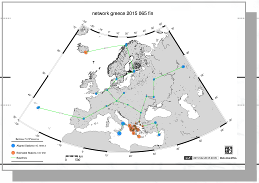
Array of Objects

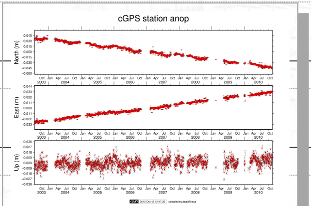
expand

8. Ambiguity Resolution SUmmary

Array of Objects



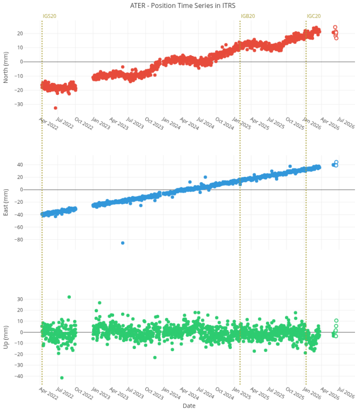




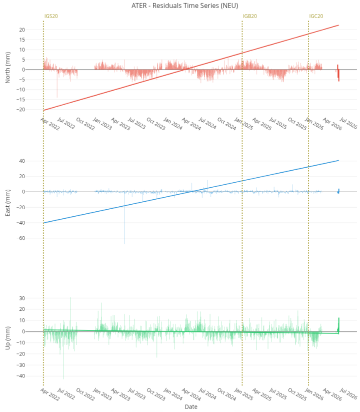
Baseline	sta1	sta2	length (km)	Method	N. of Amb.	Perce[illegible]tance	Satellite system
AUKL	AUT1	KLOK	139.7	pbnl	74	54.1	GPS
AULE	AUT1	LEMN	199.6	pbnl	60	55	GPS
KCTL	KATC	TILO	59	pbnl	50	90	GPS
KLRL	KLOK	RLSO	174.2	pbnl	74	41.9	GPS

Online time-series products

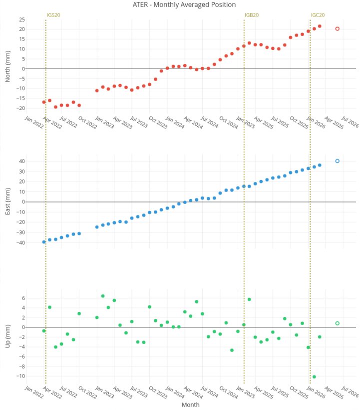
Times-series



Linear regression residuals



Monthly average



Time series model

Linear+Seasonal+PSD Terms

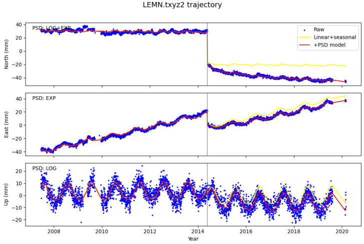
$$y(\delta t) = A\delta t + +B + C_1 \sin(\omega_1 \delta t) + D_1 \cos(\omega_1 \delta t) + C_2 \sin(\omega_2 \delta t) + D_2 \cos(\omega_2 \delta t)$$
$$+ \sum_{k=1}^{n} J_k H(t - t_k) + \sum_{i=1}^{n_l} A_i^l \log\left(1 + \frac{\delta t}{\tau_i^l}\right) + \sum_{j=1}^{n_e} A_j^e \left(1 - e^{-\frac{\delta t}{\tau_j^e}}\right)$$

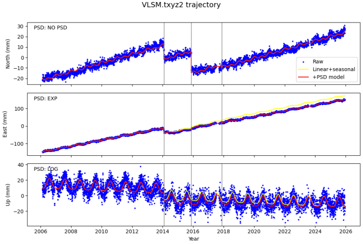
δt $(= t - t_0)$: time elapsed since the central epoch t_0 [yr]
A : station velocity [m/yr]
B : position intercept at t_0 [m]
C_k, D_k : sine and cosine amplitudes at angular frequency ω_k[m], with $\omega_1 = 2rad/yr$ (annual) and $\omega_2 = 4rad/yr$ (semi-annual)
J_k : magnitude of the k-th step offset [m], modeled by the Heaviside function $H(tt_k)$ activating at epoch t_k; ν is the total number of offsets
A_i^l, A_j^e : amplitudes of the i-th logarithmic and j-th exponential PSD terms [m]
τ_i^l, τ_j^e : relaxation time constants of the i-th logarithmic and j-th exponential PSD terms [yr]
Δt_i $(= tt_i^e q > 0)$: time elapsed since the i-th earthquake epoch $t_i^e q$; all PSD terms are zero before the earthquake
n_l, n_e : number of logarithmic and exponential PSD terms

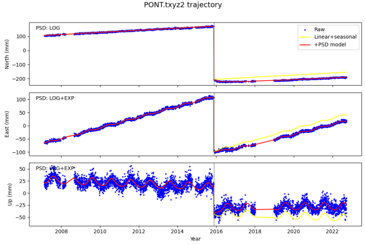
(Altamimi et al., 2023)

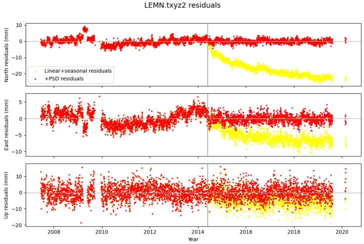
Impact of PSD modelling on Greek GNSS stations

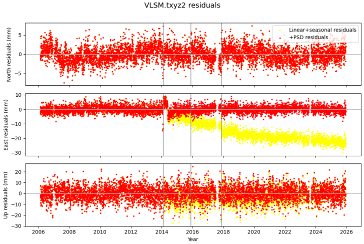
Preliminary results

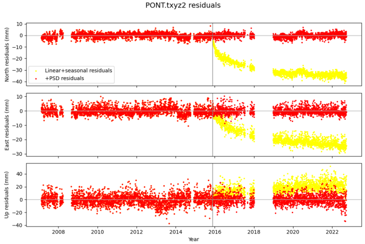




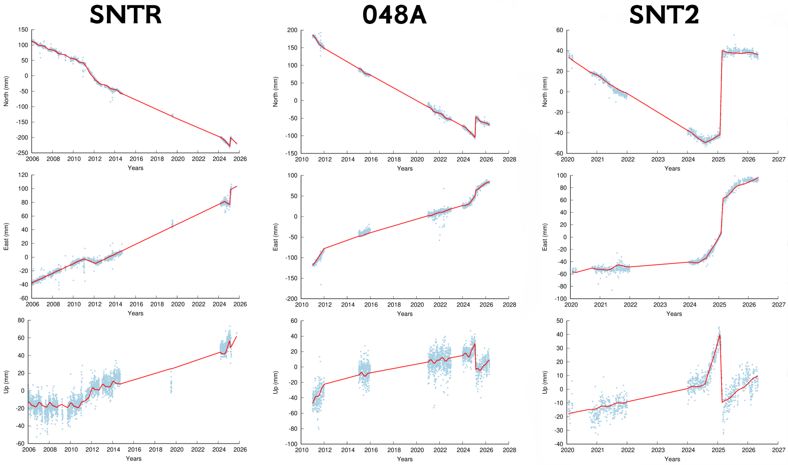




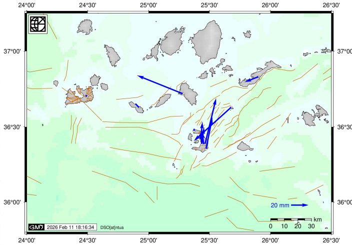




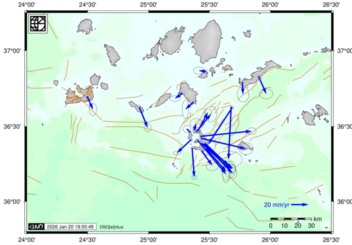
Complex station displacements - Santorini Island



Horizontal displacements 20250201 - 20250228 [IGS20]



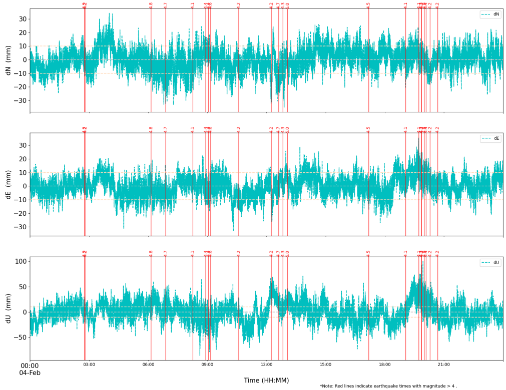
Horizontal velocities 20250303 - 20250808 [IGS20]



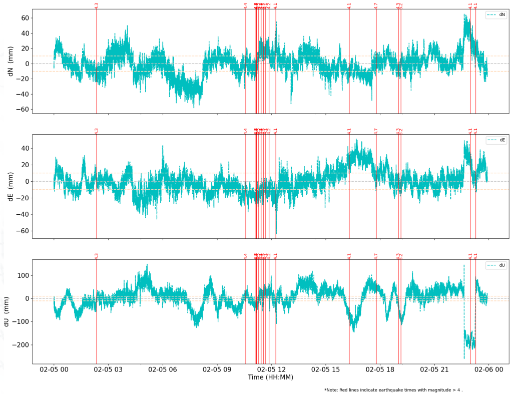
PPP processing for High-rate data

Santorini Volcano crisis (02.2025)

048A_2025_035



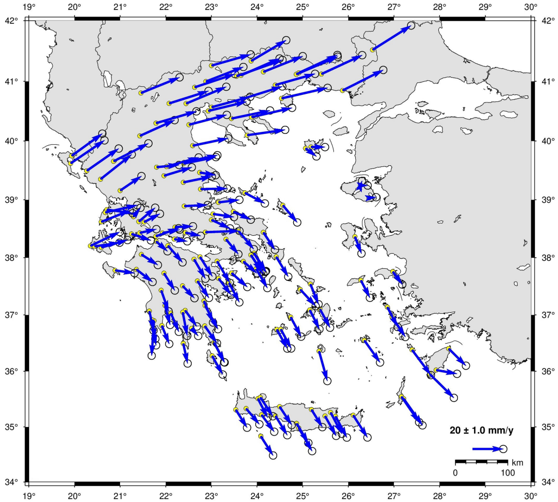
048A_2025_036



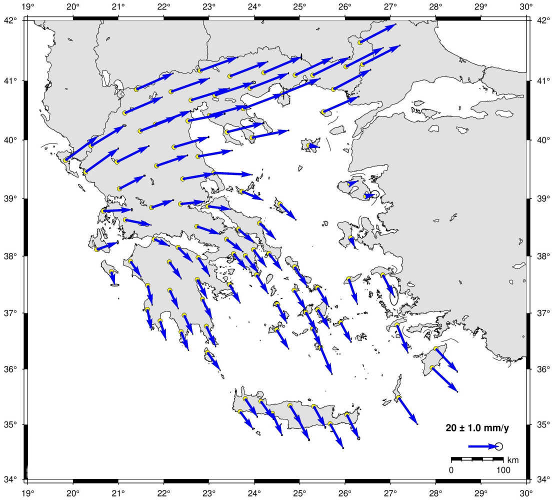
- 1Hz data available, ∼5 stations from HEPOS Network.
- PPP-AR analysis for 2 days (DOY: 035, 036)

REPRO2026: Preliminary velocity fields and network comparison

DSO_GRC_2026 (2020-2026)



DSO_HPS_2023



Conclusions and Future work

- Greece's complex tectonic and volcanic setting requires continuous and reliable GNSS monitoring.
- Routine GNSS processing provides consistent time series, station velocities, and deformation products for Geoscience applications.
- The upgraded DSO platform automates data handling, processing, database storage, time-series analysis, and visualization.
- Including post-seismic deformation (PSD) terms in time-series analysis is essential to separate transient signals from long-term tectonic velocities and avoid biased deformation estimates.
- Dense GNSS networks improve the realization of a stable local reference frame and help detect earthquakes, volcanic signals, and other transient phenomena.
- Future work focuses on expanding the datasets, supporting rapid and high-rate solutions, and strengthening DSO's contribution to the GNSS/EUREF community.

Thank you for your attention!

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